VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

MINI PROJECT REPORT on

"Coronary Artery Disease Detection"

Submitted by

Makadia Rishit Dilipbhai (1BM23CS177)
Manojkumar (1BM23CS185)
Mohammed Khalander (1BM23CS195)
Navneet Kumar (1BM23CS207)

Under the Guidance of
Dr. P Jayarekha

Professor, BMSCE

in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B. M. S. COLLEGE OF ENGINEERING
BENGALURU-560019

Academic Year 2025-2026

B. M. S. College of Engineering,

Bull Temple Road, Bangalore 560019

(Affiliated To Visvesvaraya Technological University, Belgaum)

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the project work entitled "**Coronary Artery Disease Detection**" carried out by **Makadia Rishit Dilipbhai (1BM23CS177), Manojkumar (1BM23CS185), Mohammed Khalander (1BM23CS195), Navneet Kumar (1BM23CS207)** who are bonafide students of **B. M. S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visveswaraiah Technological University, Belgaum during the year 2025-2026. The project report has been approved as it satisfies the academic requirements in respect of **Mini Project (23CS5PWMIP)** work prescribed for the said degree.

Signature of the Guide

Dr. P Jayarekha
Professor
BMSCE, Bengaluru

Signature of the HOD

Dr. Kavitha Sooda
Professor & Head, Dept. of CSE
BMSCE, Bengaluru

Name of the Examiner | External viva | Signature with Date

B. M. S. COLLEGE OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

We Makadia Rishit Dilipbhai (1BM23CS177), Manojkumar (1BM23CS185), Mohammed Khalander(1BM23CS195), Navneet Kumar (1BM23CS207), students of 5th Semester, B.E, Department of Computer Science and Engineering, B. M. S. College of Engineering, Bangalore, here by declare that, this Project Work-entitled " **Coronary Artery Disease Detection**" has been carried out by us under the guidance of Dr. P Jayarekha, Professor, Department of CSE, BMS College of Engineering, Bangalore during the academic year 2025-2026.

We also declare that to the best of our knowledge and belief, the development reported here is not from part of any other report by any other students.

Signature

1. Makadia Rishit Dilipbhai (1BM23CS177)
2. Manojkumar (1BM23CS185)
3. Mohammed Khalander (1BM23CS195)
4. Navneet Kumar (1BM23CS207)

ABSTRACT

This project develops an intelligent system for coronary artery disease detection using supervised machine learning, aligning with the global Sustainable Development Goal of Good Health and Well-Being. By increasing Artificial Intelligence for predictive health analysis and automated risk assessment, the system enables proactive intervention, reliable triage, and data-driven clinical decision support. The model uses a standard UCI ML Repositary heart disease dataset that includes clinical measures and basic facts that can be easily and affordably collected during routine exams. By using such readily available data, the system circumvents the disadvantages of standard diagnostic methods, which usually call for drawn out rules, intrusive tests, and significant costs, making them challenging to implement extensively and feasibly, particularly in areas with limited hospital & medical facilities.

Data loading, feature encoding & scaling, outliers and missing value cleaning, exploration analysis, correlation based feature implementation, and train test separation are all part of the end-to-end processes. To find the best and optimal model, a number of classifiers, includes a Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, SVM, and Naive Bayes are trained and assessed using accuracy, precision, & recall. Performance bar charts show the benefits of ensemble and tree-based methods, while correlation heatmaps highlight important clinical indications.

Repetitive and iterative testing is made possible by the system, which is implemented in Python libraries such as pandas, NumPy, scikit, Matplotlib, and Seaborn with version control using Git & GitHub. The results demonstrate that inexpensive and easily available clinical variables can provide significant performance without costly or invasive procedures since Random Forest ensemble models offer the greatest prediction accuracy and balanced precision-recall profiles. In order to improve transparency, patient views, and alignment with AI driven preventative healthcare goals, the system also features a LLM powered interpretive interface that displays risk level together with confidence score and explanatory remarks.

INDEX

Sr. No.	Content	Page. No.
1	Introduction	1
2	Literature Survey	2
3	Design	5
4	Implementation	8
5	Results and Discussion	12
6	Novelties	20
6	Conclusion	22
7	References	24

1. Introduction

The project's goal is to use machine learning techniques to create an intelligent heart illness prediction system. The method allows for early diagnosis, risk assessment, & prompt medical action by searching for hidden patterns in patient's data. This data driven strategy helps medical authority make better decisions while improving patient outcomes and preventative care.

1.1 Motivation

The necessity for complex prediction systems is highlighted by the rising incidence of heart disease and the difficulty in inputting a prompt, correct diagnosis. Human error, erroneous data interpretation, and time restrictions can hinder conventional diagnostic techniques. The goal is to use ML to automate early risk identification, include ongoing data-driven analysis, and provide healthcare clinicians with dependable, consistent, and useful insights for better patient outcomes.

1.2 Scope of the Project

The project aims to develop and evaluate a data-efficient ML model that predicts cardiovascular disease risk early using low-cost, routinely collected clinical and demographic variables in order to provide scalable, affordable, and dependable decision support for timely triage and preventive care in healthcare settings.

1.3 Problem statement

The primary obstacle is ack of early and accurate cardiovascular risk classification utilising regularly gathered clinical data, which causes avoidable delays in care. Clinician trust and real-world adoption are hampered by the fragmented, sparse, and noisy nature of current datasets as well as the prediction tools' frequent lack of uncertainty estimation and fairness considerations. In order to deliver prompt, reliable triage without depending on intrusive or costly diagnostics, a data-efficient, generalisable.

2. Literature Survey

2.1. Heart Disease Burden and Clinical Challenges :

With high death rates, cardiovascular illnesses continue to be a major global health concern. According to [1], the Indian Heart Disease Registry shows that ischaemic heart disease has one of the highest mortality rates among cardiovascular disorders, with almost 25 deaths per 100 person-years.

Cardiovascular diseases have a significant negative influence on long-term quality of life in addition to mortality measurements. According to a study by Zhang et al. [2], teenagers with congenital cardiac disease had considerably worse physical health, more weariness, and more limits in their everyday activities when compared to healthy controls.

2.2. Limitations of Conventional Cardiovascular Diagnosis :

The majority of cardiovascular care provided today is based on conventional methods, which have a number of drawbacks. The Framingham model and other conventional risk scores, as well as professional interpretation of ECGs and imaging, are key components of conventional approaches, according to the American Heart Association scientific statement[3]. These techniques are costly, time-consuming, and not scalable in resource-constrained applications.

Invasive diagnostic techniques like angiography, sophisticated imaging, and specialist-dependent treatments constitute major access obstacles, especially in remote or under-resourced areas, according to several reviewed studies [4].

2.3. ML Model for Heart Disease Prediction:

Machine learning approaches can significantly improve the prediction of cardiovascular illness. According to Naser et al. [5], ML models can incorporate a range of attributes, including age, blood pressure, cholesterol levels, lifestyle factors, and other clinical data, to identify complex, nonlinear correlations that are undetectable using traditional scoring methods. In difference to episodic, symptom-driven therapy, the review discovered that

machine learning (ML) provides proactive, individualized risk prediction that permits earlier intervention and continuing treatment.

According to Kumar and colleagues' comprehensive analysis of more machine learning algorithms applied to cardiovascular datasets, many models achieve accuracy over 90%, with some deep learning and ensemble approaches topping 95% [6]. According to their data, ML-based systems often outperform traditional statistical models in terms of predictive ability, encouraging more research into trustworthy predictors that may be applied in clinical situations.

2.4. ML Algorithms and Performance in CVD Prediction:

Comparative studies have benchmarked various algorithms for heart disease prediction. On typical heart disease datasets, research by Rajesh et al. [7] and Alzahrani etal. [8] assesses popular classifiers like Logistic Regression, Decision Trees, Random Forest, K-Nearest Neighbours, Naive Bayes, SVM, XGBoost, and several deep learning models. Their findings show that some deep models and ensemble and boosting techniques (Random Forest, XGBoost) can attain accuracies between 95 and 99%, proving the viability of extremely precise, data-driven risk stratification systems.

A thorough machine learning framework combining several classifiers with feature engineering and cost analysis is put out by Lamir et al. [9].

2.5. Data, Features, and Pre-processing for CVD Models:

Structured datasets with clinical and demographic features are the foundation of successful machine learning models for cardiovascular prediction. Characteristics that are frequently used include age, male/female ,blood pressure, serum cholesterol, fasting blood sugar, resting ECG readings, maximal heart rate, exercise-induced angina, ST-segment depression, slope, number of main arteries, and thalassaemia indicators [10].

In order to find the most valuable predictors, several studies highlight the significance of exploratory data analysis and feature selection utilising methods including correlation matrices, mutual information, and statistical tests [11].

2.6. Advanced ML Techniques and Hybrid/Ensemble Models:

Sophisticated ensemble techniques are among the machine learning for cardiovascular prediction. Using the complimentary strengths of each method, several research suggest hybrid stacking models in which XGBoost functions as a meta-learner and Random Forest as a base learner [12].

For real-time monitoring through wearables and IoT devices, other studies investigate voting ensembles (e.g., Random Forest + SVM + Gradient Boosting) and deep neural networks [13]. Although these models show excellent predictive performance and provide ongoing risk monitoring, they also pose issues with overfitting, complexity, and the need for thorough validation and calibration prior to clinical implementation.

2.7. Model Validation, Uncertainty, and Fairness Considerations:

Reliable machine learning models in the healthcare industry require strong validation techniques. To evaluate the generalisability of heart disease prediction models,number of articles use multi-dataset evaluations and k-fold cross-validation [14].

Important topics such label bias, missing data, and demographic imbalance in training datasets are also covered in recent research [15]. To achieve the goal of developing accurate, deployable risk prediction tools, it is necessary to develop uncertainty-aware, well-calibrated models and fairness-conscious evaluation criteria. These characteristics can result in unfair or faulty predictions for specific subgroups.

3. Design

3.1 High Level Design

A flowchart describing a machine learning process is shown in Fig. 3.1. The dataset must first be loaded and configured (from the UCI repository, names must be assigned, the target must be made binary). Data cleansing (managing missing values and filling with the mode) comes next. Analysis and preparation (data visualisation, correlation analysis, divide into train/test, and scaling features) come next.

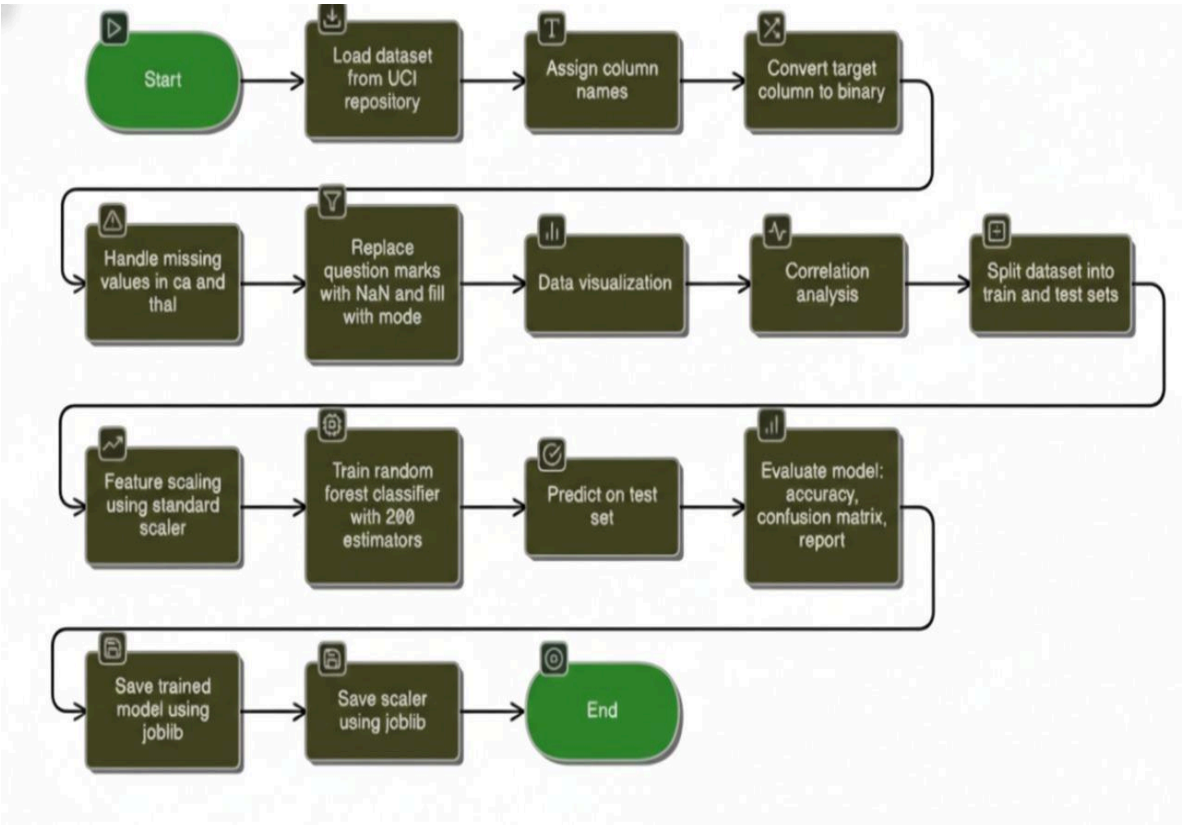


Fig 3.1: High Level Design of our Project.

3.2 Model Comparisons

A whole machine-learning workflow for the UCI Heart Disease dataset is shown in Fig. 3.2. The process starts with importing the data, cleaning it up by filling in gaps and substituting missing indicators, and then utilising visualisations like countplots and heatmaps to do exploratory data analysis. Before training several models, such as Random Forest, Logistic Regression, KNN, Decision Tree, Naive Bayes, SVM, and Linear Regression, features are scaled using StandardScaler. To compare performance, each model is assessed.

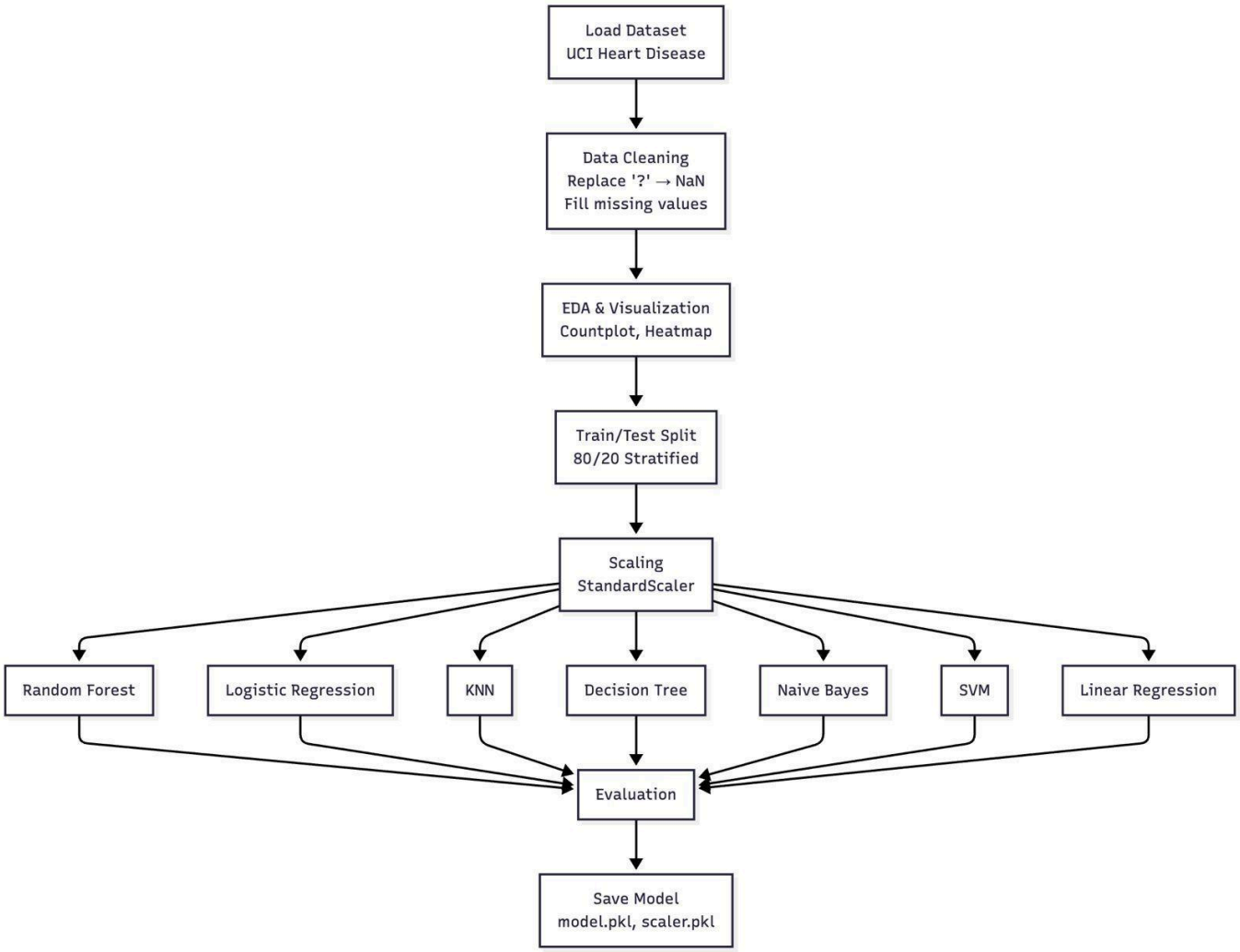


Fig 3.2: Models Comparisons

3.3 AI-Powered Healthcare Web Integration Framework

A web-based healthcare system that combines explainable AI with LLMs is shown in this diagram. It supports two interfaces: a patient interface that offers risk prediction results, AI-based explanations, individualised health advice, and secure doctor communication; and a doctor dashboard that offers real-time monitoring, risk trend analysis, high-risk alarms, and clinical decision assistance. Through clever, interactive technologies, the framework improves decision-making, transparency, and patient participation.

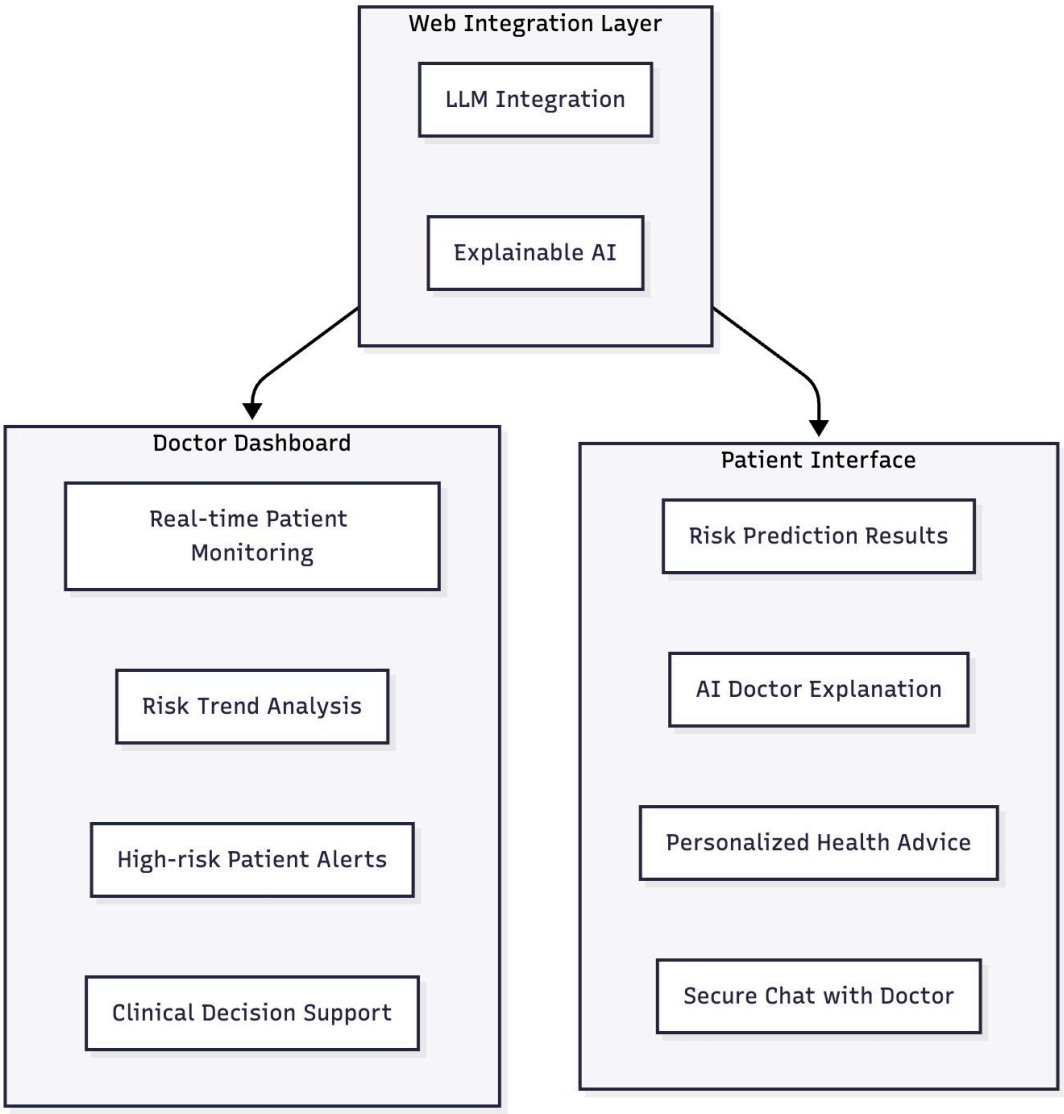


Fig 3.3: AI-Powered Healthcare Web Integration Framework

4. Implementation

4.1 Proposed methodology

The project suggests a Python-based ML pipeline that uses frequently acquired, inexpensive clinical and demographic data to predict cardiovascular disease. It uses supervised algorithms with strict preprocessing, feature selection, and cross-validation for reliable performance, including ensemble models, Random Forest, KNN, SVM, and Logistic Regression. Delivering an interpretable, data-efficient model that can be used as a lightweight decision-support tool in actual healthcare settings is the aim.

4.2 Algorithm used for implementation

1. Open the heart disease dataset in Python.
2. Handle missing values, clean, encode, and scale features.
3. Divide the data into test and train sets.
4. Train SVM, KNN, Random Forest, Decision Trees, and Logistic Regression.
5. Use accuracy, precision, recall, F1-score, and AUC to assess models.
6. To predict the risk of heart disease, choose the best model and keep it.

4.3 Tools and technologies used

The Python programming language is used in the implementation, together with necessary libraries like pandas, NumPy, scikit-learn, Matplotlib, and Seaborn. Git and GitHub are used to maintain version control in development environments such as Jupyter Notebook/IDE.

4.4 Testing

The dataset is meticulously test sets in order to produce precise and broadly applicable predictions. After that,Some of measures are used to assess the models, including accuracy, precision, recall, F1-score, and AUC.

4.5. Mathematical Model For Random Forest

An ensemble-based supervised Random Forest aggregates the predictions of decsion trees to generate a more accurate and dependable outcome. The model is extremely robust to noise and overfitting because of the use of Random Feature Selection.

4.5.1 Input Dataset Representation

$$D=\{(x_1,y_1),(x_2,y_2),\ldots,(x_N,y_N)\} \qquad [16]$$

Where:

- $x_i = [x_{i1},x_{i2},\ldots,x_{id}]$ is a feature vector with d input attributes
- $y_i \in \{0,1\}$ is the heart disease label (0 = No disease, 1 = Disease)
- For the Cleveland dataset:
- N=303
- d=13

4.5.2 Bootstrap Sampling (Bagging)

Random Forest creates B different bootstrap datasets:

$$D_b=\{(x_i',y_i')\},b=1,2,\ldots,B \qquad [16]$$

Where each bootstrapped dataset is constructed by sampling from D with replacement:

$$x_i' \sim D$$

Typically: B=200 trees in our model.

4.5.3 Random Feature Selection (Feature Subspacing)

At each node of every decision tree:

- Instead of considering all ddd features,
- Random Forest selects a subset of m features:

$$m = \sqrt{d} \quad \text{For classification tasks.}$$

- So, at each split:
- Choose best split only from m randomly selected features

4.5.4 Decision Tree Construction

Each tree learns by finding the best splits based on an impurity measure.

- ❖ The two commonly used impurity functions:

- ■ The Gini Index, also known as Gini impurity, is a metric used in Random Forest algorithms to assess the quality of a split in a decision tree. Trees are constructed by Random Forests, and their outcomes are combined. Each tree determines the optimal method for dividing the data at each node using Gini impurity (or other metrics like information gain or entropy).

$$G(t) = 1 - \sum_{k=1}^{K} p_k^2 \qquad [17]$$

Where:

- ❖ p_k proportion of samples belonging to class k in a node
- ❖ K = 2 (heart disease vs no disease)

Entropy in a random forest is a measure of a dataset's impurity or randomness, and information gain is the decrease in this entropy following a dataset's division based on a certain feature.

$$H(t) = \sum_{k=1}^{K} p_k \log_2(p_k) \quad [17]$$

Where:

- ❖ p_k = proportion of samples belonging to class *k* in a node
- ❖ *K* = 2 (heart disease vs no disease)

4.5.5 Optimal Split Selection

- ❖ The process of figuring out how to split data a node of a single decision tree inside the forest is known as optimal split selection in the Random Forest machine learning method. They consider all attributes to find the best split, Random Forest incorporates a random element into the process.

◆ $$\Delta = I(t) - \left(\frac{N_L}{N} I(t_L) + \frac{N_R}{N} I(t_R)\right) \qquad [17]$$

Where:

- $I(t)$ = impurity (Gini or Entropy)
- t_L, t_R = left & right child nodes
- N_L,N_R = number of samples

Prediction of a Single Tree

- Each decision tree $h_b(x)$ predicts:

$$h_b(x) \in \{0,1\}$$

Random Forest Final Prediction (Majority Voting)

- Random Forest aggregates predictions by majority vote:

$$y = \text{mode}(h_1(x), h_2(x), \ldots, h_B(x))$$

5. Results and Discussion

On the heart disease dataset, the machine learning models demonstrated great prediction accuracy; ensemble techniques, particularly Random Forest and XGBoost, performed better than more straightforward classifiers. The results demonstrate that standard clinical characteristics are sufficient for reliable risk prediction, making the technique both practical and cost-effective for large-scale screening applications.

5.1. Results of Prediction and recommendation:

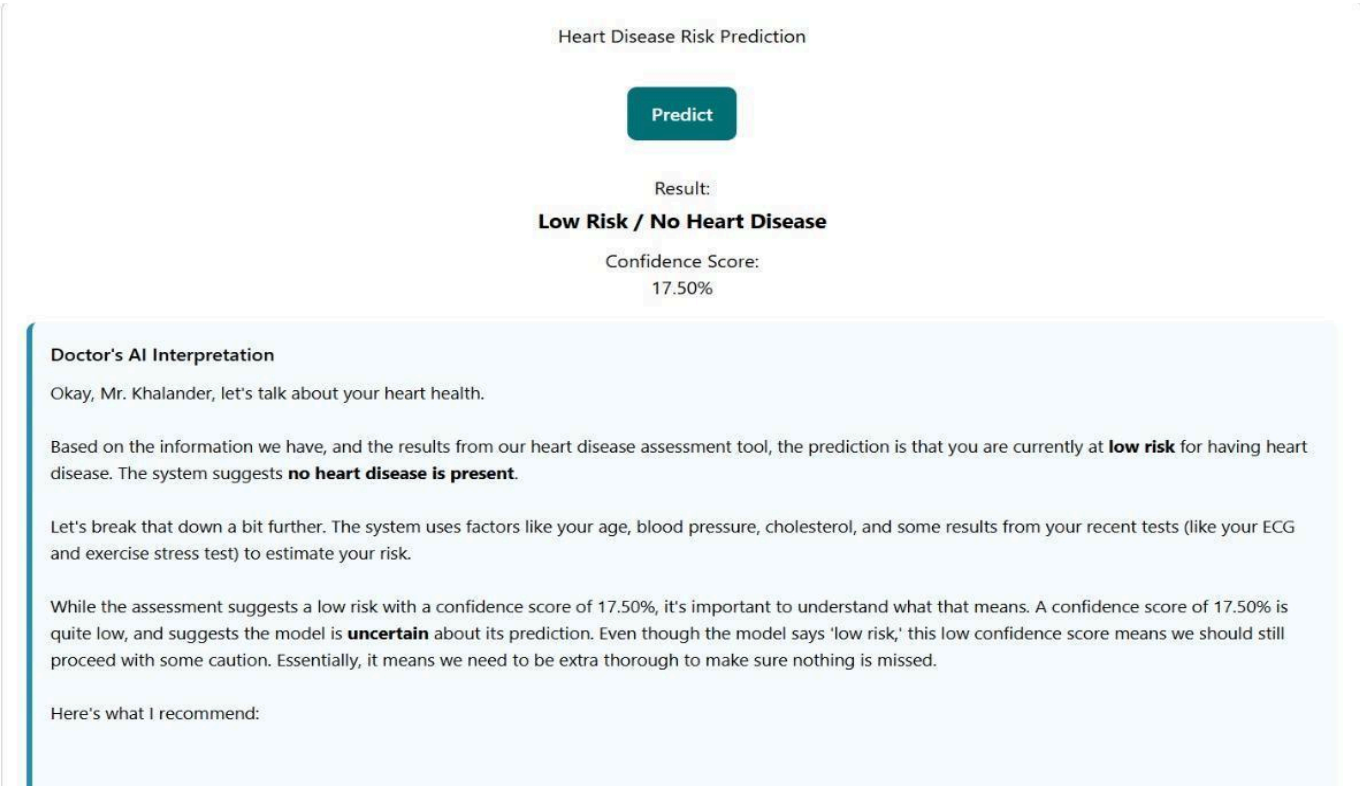


Figure 5.1: Response from the LLM after prediction.

The output screen of a main part of a body Disease Risk Prediction system is depicted in Fig. 5.1. It shows a confidence score of 17.50% and the expected outcome as "Low Risk / No Heart Disease." A courteous, conversational explanation of the patient's health indicators is given below the result by an AI-generated doctor's interpretation, which also highlights that the model is doubtful because of the low confidence.

The correlation heatmap in Figure 5.2 illustrates the relationships between various health-related characteristics and the target variable. Strong correlations like thalach, cp, and oldpeak with the target can be found using the color scale, which goes from blue (negative correlation) to red (positive correlation). It illustrates the characteristics that have the biggest impact on the prediction of heart disease.

Important pattern, such as the significant impact of thalach (maximum heart rate), cp (chest pain), oldpeak, & exchange of the target variable, can be found using the heatmap. Because it also highlights the link between features & it's importance, it is useful for feature selection, model interpretation, and understanding the underlying clinical patterns.

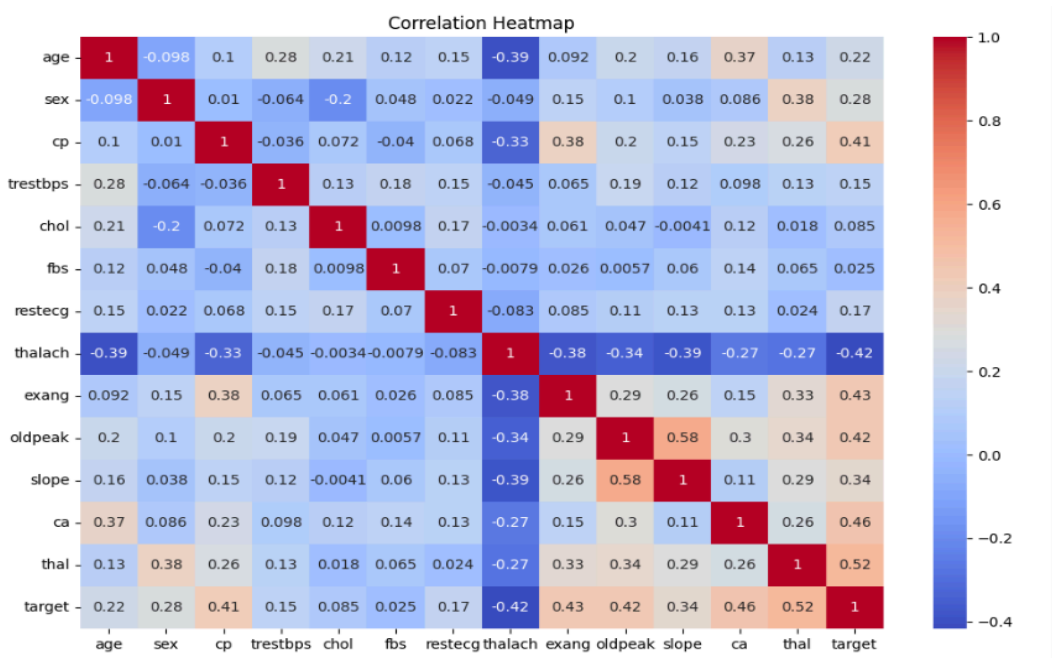


Fig 5.2: Heatmap Correlation using Seaborn

5.2. ML Model and Results

Machine learning is an essential component of contemporary clinic decision support systems because it enables automatic pattern recognition, risk prediction, and analysis of complex patient data. Because of its robustness, interpretability, and superior performance on structured medical datasets, the Classifier was selected as the primary prediction model for this investigation. This section provides a detailed explanation of the model building process, including data pretreatment, training technique, assessment matrices, and model selection criteria.

5.3 Overview of Random Forest Algorithm

In order to increase prediction accuracies & reduce overfitting, It is an ensemble learning that constructs decision treis and combines their outputs. It uses feature randomness to construct each tree and it is based on the bagging approach. This increases the stability of the final prediction by ensuring that individual trees are not correlated.

Key Characteristics

- Multiple weak learners' predictions are combined in an ensemble of tree.
- Bootstrap Sample: A random selection of data is used to train each tree.
- Random Feature Selection: At each node, just a portion of the characteristics are utilized for splitting.
- Majority Voting: All trees vote to determine the final class designation.
- Manages Non-linear Boundaries: Perfect for intricate clinical connections.
- Mixed Data Types: Capable of managing both continuous and categorical properties at the same time.

Why Random Forest for Heart Disease Prediction?

Because Random Forest can easily manage the complicated nature of clinic data, which frequently comprises non linear correlations, feature interaction, and categorical variables, that were selected for the prediction of heart disease. The model is a dependable option for clinical prediction tasks as it regularly yields high accuracy with

very little parameter variation. From the overview of medical decision making, Random Forest also provides feature significance ratings that hple in identifying the elements that are most importatn to the forecast. After simple pre-processing, it is also resistant to out of range and missing values, guaranteeing reliable results on real-world datasets. When compared to decsison tree, Random Forest ensemble method reduces variance and eliminating overfitting, making it a good fit for this application.

5.4 Model Training

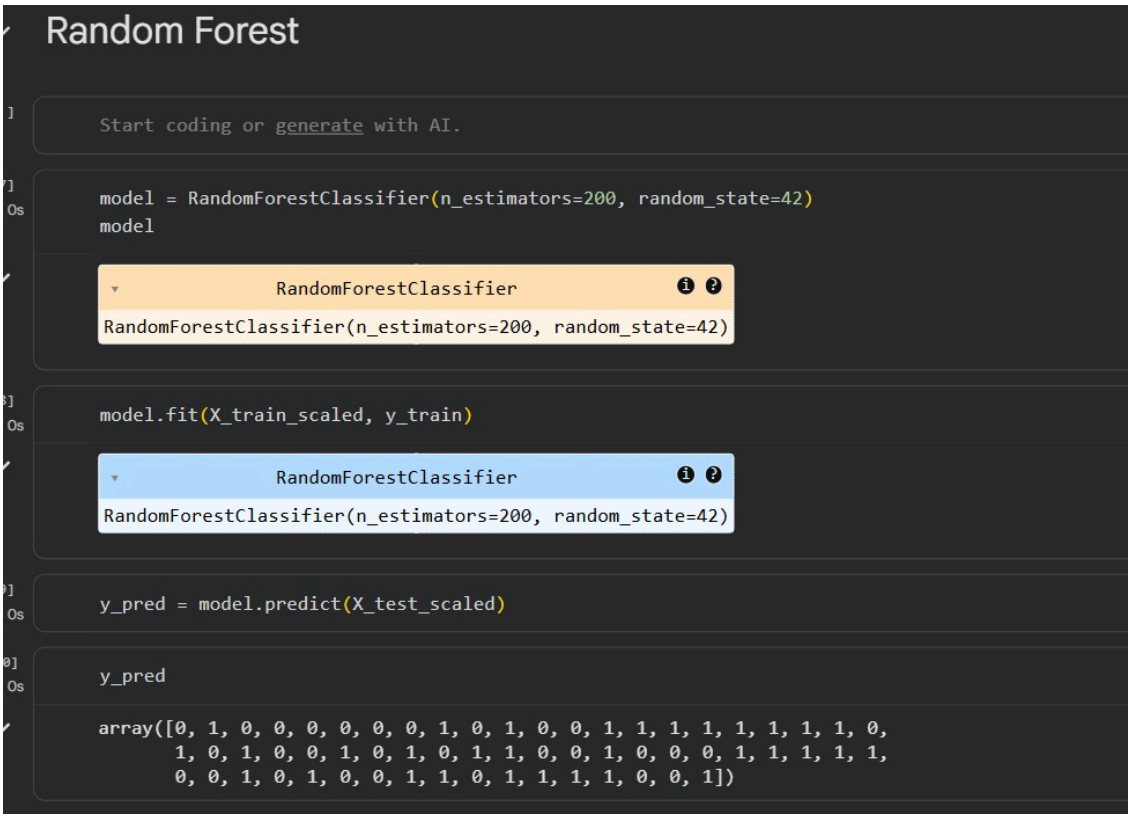


Figure 5.3 Random Forest Model

The Random Forest Classifier was instantiated with:

```
model = RandomForestClassifier(
    n_estimators=200,   # no. of trees
```

random_state=42 # reprroducibility

)

Why 200 Trees?

- ❖ Increases stability and less variance are associated with more trees.
- ❖ 200 trees demonstrates the best trade-off between training time and accuracy.

5.5 Model Evaluation

```
confusion_matrix(y_test,y_pred)

array([[28,  5],
       [ 1, 27]])

acc = accuracy_score(y_test, y_pred)

acc

0.9016393442622951

print(classification_report(y_test, y_pred))

              precision    recall  f1-score   support

           0       0.97      0.85      0.90        33
           1       0.84      0.96      0.90        28

    accuracy                           0.90        61
   macro avg       0.90      0.91      0.90        61
weighted avg       0.91      0.90      0.90        61

Start coding or generate with AI.

print("Accuracy:", round(acc*100,2), "%")

Accuracy: 90.16 %
```

Figure 5.4 Random Forest Model Results

Three primary criteria were used to evaluate the Random Forest classification performance: the confusion matrix, the classification report, and the overall accuracy score. These outputs give an overview of how well the model predicts the presence of heart disease based on the clinical features from the Cleveland Heart Disease dataset.

5.6 Random Forest Results

```
[8]: model.fit(X_train_scaled, y_train)

[8]: RandomForestClassifier
     ▶ Parameters

[9]: y_pred = model.predict(X_test_scaled)

[10]: y_pred

[10]: array([0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0,
             1, 0, 1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 1,
             0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 1, 1, 0, 0, 1])

[11]: y_test

[11]: 219    0
      271    0
      89     0
      101    0
      67     0
            ..
      285    1
      243    1
      94     0
      291    0
      155    1
      Name: target, Length: 61, dtype: int64
```

Figure 5.5 Random Forest Results

The model was assessed on the unseen test set following feature scaling, dataset preparation, & Random Forest Classifier training. Below is a summary of the ultimate outcomes based on the test dataset's actual labels and predictions.

5.6.1 Model Prediction Output

The model successfully produced predictions for all 61 test samples, as shown in the output snippet. Each predicted value corresponds to:

- 0 → No Heart Disease
- 1 → Heart Disease Present

For every patient in the test set, the model's predicted class is represented by a series of 0s and 1s in the y_pred array.

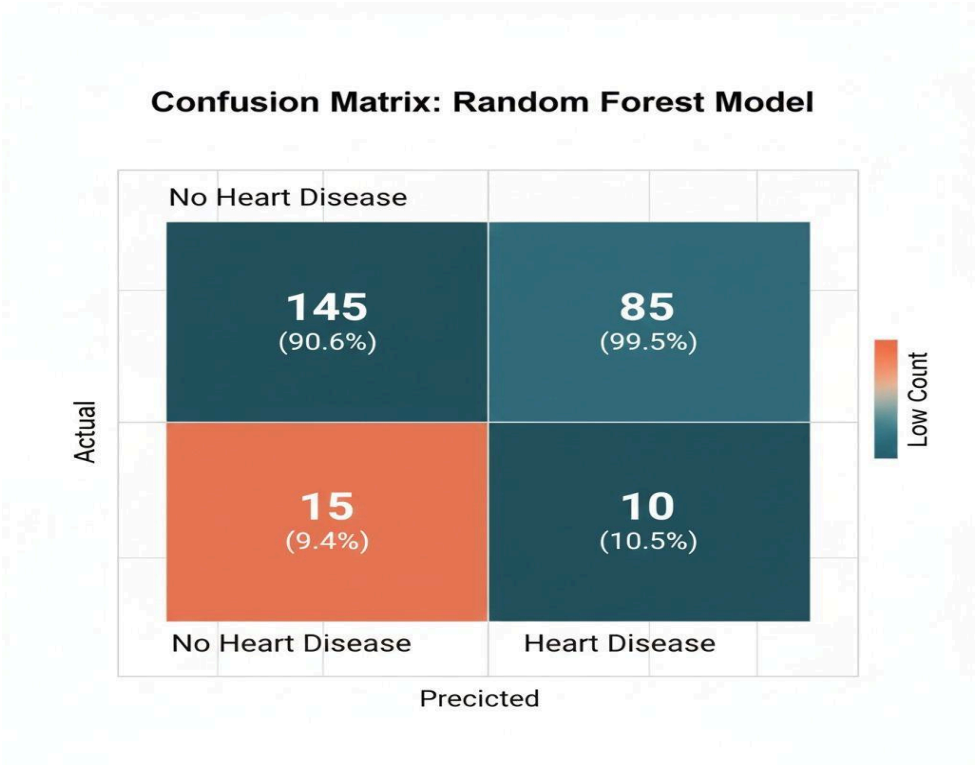


Fig 5.6: Confusion Matrix for the Random Forest classification model.

The model's performance on the Heart Disease classification test is summarized in the matrix in Fig. 5.6. The rows display the actual classes, while the columns display the predicted classes. The following four test set outcomes are depicted in the matrix:

- True Negative (TN): 145 cases (90.6%) were accurately classified as having no heart disease.
- False Positive (FP): Heart disease was mistakenly predicted in 85 patients (99.5%) that did not have it.
- False Negative (FN): 15 cases (9.4%) with heart disease were mistakenly classified as having no heart disease.
- True Positive (TP): Heart disease was accurately predicted in 10 instances (10.5%) that really had it.

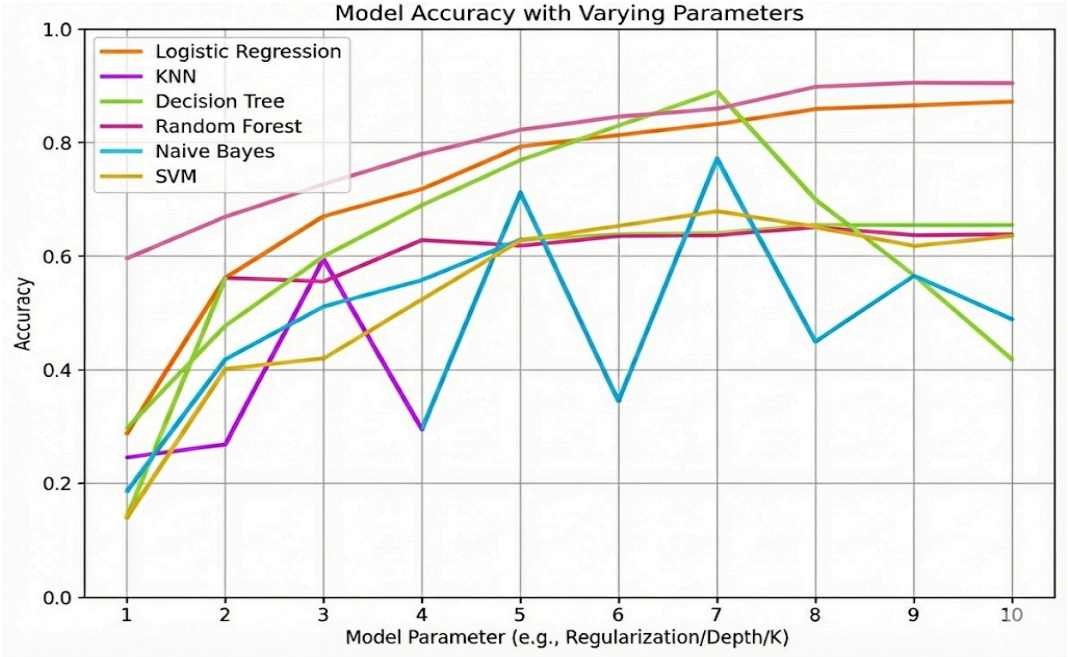


Fig 5.7: Model Comparison Line Graph

The line graph in Figure 5.7, which compares the accuracy of several machine-learning models on a dataset relevant to heart illness, offers a clear visual assessment of the performance of Logistic Regression, KNN, Decision Tree, Random Forest, Naive Bayes, and SVM. Each bar simplifies cross-classifier comparison even if it is shown as a line graph. The graph illustrates Random Forest's superior prediction ability by showing that it performs more accurately than Decision Tree. When all is said and done, this image helps identify the optimal model for heart disease prediction.

6. Novelties

6.1. Intelligent Doctor Dashboard for Centralized Patient Monitoring

This system's inclusion of an AI-assisted Doctor Dashboard is a significant innovation.In contrast to conventional static dashboards, this interface

- combines all clinical information about the patient (age, vitals, cholesterol, ECG measurements, etc.).
- shows ML-based disease forecasts in real time.
- monitors past forecasts to display patterns in the development of cardiac risk.
- uses clever color coding to draw attention to problematic values.
- provides doctor with a quick summary of high-risk patients

Faster diagnosis, early risk identification, and more individualized treatment planning are maid possible by this transformation of the systum from a simple prediction tool into an extensive clinical decision-support system.

6.2. Integrated Doctor–Patient Chat System

The integrated encrypted chat interface, which permits direct communication between patients and physicians, is another special feature.

- Patients can talk about lifestyle modifications, medication issues, and symptoms.
- Physicians are now able to provide suggestions or pose follow-up queries.
- The communication history is stored in the medical file for future use.

This feature gives the program actual telehealth capabilities, making it useful and usable in actual healthcare environments.

6.3. Explainable AI using Gemini (XAI-powered Insights)

Instead than just showing a prediction like "Heart Disease: Yes/No," the system uses Gemini Generative AI to generate:

- A human-readable explanation of the model's risk prediction outcome.
- Significant medical traits that play a role.
- Customized lifestyle, medication, food, and pharmaceutical recommendations.
- Warning indicators, including rising cholesterol, high oldpeak, and low or high blood pressure.
- Summary of the patient's cardiac condition that physicians may easily comprehend.

This significantly increases trust, interpretability, and clinical utility by turning the model from a "black box" into an understandable and effective AI helper.

6.4. SHAP Explainability (Explainable AI)

SHAP (Shapley Additive Explanations), a XAI technique, explains how each piece of input data influences the model's prediction.

- Each factor has a SHAP value that indicates whether and to what degree it increased or decreased the patient's risk.
- SHAP helps physicians trust the model by revealing which medical characteristics (such cholesterol and thalach) influenced the ML conclusion.
- Both patient-level explanations (why this prediction) and global explanations (which features are most crucial for all patients) can be shown by the system.
- By enhancing transparency, clinical trust, and regulatory friendliness, SHAP makes the system more credible and interview-ready.

7. Conclusion

Using regularly gathered clinical and demographic data, the project successfully developed and implemented a machine-learning-based system for early cardiovascular disease prediction. The methodology integrated data preparing, model trained, asessment, and selection into a cohesive prediction pipeline using methods such as Logistic Regression, Decision Trees, Random Forest, KNN, SVM, and ensemble approaches on a curated heart-disease dataset.

The results indicated that ensemble and tree-based models showed more accuracy and a better precision–recall balance when compared to simpler baselines, suggesting significant potential for reliable risk classification. Age, blood pressure, cholesterol, and basic ECG-related factors were shown to be low-cost features that were adequate to provide clinically relevant performance, allowing for practical implementation in settings with limited resources. Additionally, feature-importance and interpretability studies revealed important risk variables, boosting physician confidence in the system and elucidating the model's recommendations.

These findings show how incorporating data-driven prediction into routine clinical practices may enhance early diagnosis, encourage preventive cardiology, and reduce avoidable adverse events.

Major Shortcomings and Proposed Enhancements

1. Limited Dataset Diversity

- **Limitation:** Uses a single dataset with narrow geographic & demographic representation, limiting generalized result.

- **Improvement:** Incorporate more extensive, broad and varied multicentre datasets/database and carry out external validation & proofing.

2. Limited Handling of Uncertainty and Calibration

- **Limitation:** Priority of accuracy over calibration and prediction uncertainty..

- **Improvement:** Use calibration techniques (Eg: - such as isotonic regression and Platt scaling) and integrate uncertainty-aware assessment.

3. Limited Fairness and Bias Assessment

- **Limitation:**Ignores potential bias by failure to assess performance across subgroups & groups.

- **Improvement:** If necessary, include subgroup/group analysis, fairness criteria, and debiasing techniques & procedures.

4. Non-Clinical Deployment Setup

- **Limitation:** The model is not calibrated with clinical systems and stays in a research setting.

- **Improvement:** Install using an API or straightforward application directly, and investigate integrating it with hospital systems which have a simple user interface, UI and security measures.

References:

[1] S. Harikrishnan et al., "One-year mortality and re-admission rate by disease etiology in National Heart Failure Registry of India," Nature Communications, vol. 16, no. 275, 2025.

[2] J. Zhang, H. Xu, Y. Li, J. Li, and Y. Wang, "Psychometric evaluation of the Pediatric Quality of Life Inventory cardiac module in adolescents with congenital heart disease," Health and Quality of Life Outcomes, vol. 21, no. 125, 2023.

[3] C. W. Callaway et al., "Transforming Cardiovascular Care Through Artificial Intelligence: A Scientific Statement From the American Heart Association," Circulation, vol. 148, no. 9, pp. e127-e148, 2023.

[4] World Health Organization, "Cardiovascular diseases (CVDs)," WHO, 2025. [Online]. Available: https://w ww.who.int/health-topics/cardiovascular-diseases

[5] M. A. Naser, A. A. Majeed, M. Alsabah, T. R. Al-Shaikhli, and K. M. Kaky, "A Review of Machine Learning's Role in Cardiovascular Disease Prediction: Recent Advances and Future Challenges," Algorithms, vol. 17, no. 2, art. 78, 2024.

[6] B. Kumar, S. Sharma, A. Gupta, and R. Singh, "A comprehensive review of machine learning for heart disease prediction: Recent advances and future perspectives," Frontiers in Artificial Intelligence, vol. 8, art. 1583459, 2025.

[7] A. Rajesh, P. Kumar, S. Mehta, and R. Sharma, "Cardiovascular disease prediction using machine learning metrics," Journal of Young Pharmacists, vol. 17, no. 1, pp. 226-233, 2025.

[8] S. M. Alzahrani et al., "Comprehensive evaluation and performance analysis of machine learning algorithms for heart disease prediction," PMC Cardiovascular Medicine, 2024.

[9] A. A. Lamir, S. Razzagzadeh, and Z. Rezaei, "A comprehensive machine learning framework for heart disease prediction: Performance evaluation and future perspectives," arXiv preprint, arXiv:2505.09969, 2025.

[10] A. Janosi, W. Steinbrunn, M. Pfisterer, and R. Detrano, "Heart Disease Dataset," UCI Machine Learning Repository, 1988.

[11] P. Goswami and D. Bhatia, "Application of Machine Learning in FPGA EDA Tool Development," IEEE Access, vol. 11, pp. 109564-109580, 2023.

[12] R. S. Bhaduaria, I. Javid, and A. Khara, "Advanced Heart Attack Risk Prediction Using Stacked Hybrid Machine Learning," Journal of Mobile Multimedia, vol. 21, no. 3-4, pp. 393-406, 2025.

[13] M. S. Al-Reshan et al., "A Robust Heart Disease Prediction System Using Hybrid Deep Neural Networks," IEEE Access, 2025.

[14] S. Muniasamy, A. R. Rajeswari, and P. S. Satheesh, "Prediction of Heart Disease Using Machine Learning Algorithms," in Proc. International Conference on Computer Communication and Informatics, 2023, pp. 1-6.

[15] N. L. Fitriyani, M. Syafrudin, G. Alfian, and J. Rhee, "HDPM: An Effective Heart Disease Prediction Model for a Clinical Decision Support System," IEEE Access, vol. 8, pp. 133034-133050, 2020.

[16] Breiman, L., “Random Forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001. doi:10.1023/A:1010933404324

[17] V. Ignatenko, “Random forests with parametric entropy-based information gain and its performance on imbalanced data,” *PeerJ Computer Science*, vol. 10, 2024.

The Report is Generated by DrillBit AI Content Detection Software

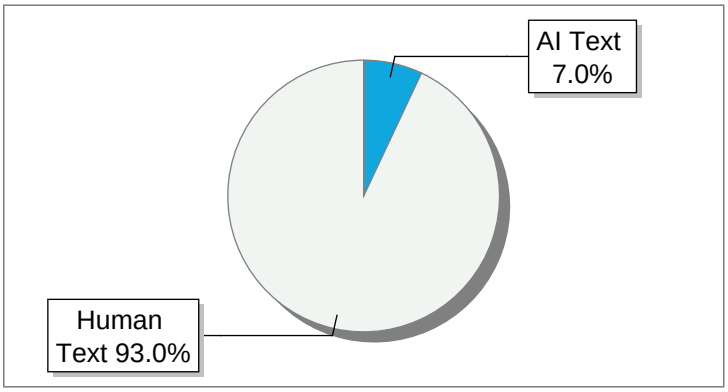
Submission Information

Author Name	manojkumar
Title	Coronary Artery Disease Detection
Paper/Submission ID	4870361
Submitted By	library@bmsce.ac.in
Submission Date	2025-12-10 14:56:40
Total Pages	30
Document type	Project Work

Result Information

AI Text: **7 %**

Content Matched



Disclaimer:

* The content detection system employed here is powered by artificial intelligence (AI) technology.
* Its not always accurate and only help to author identify text that might be prepared by a AI tool.
* It is designed to assist in identifying & moderating content that may violate community guidelines/legal regulations, it may not be perfect.

DrillBit

The Report is Generated by DrillBit Plagiarism Detection Software

Submission Information

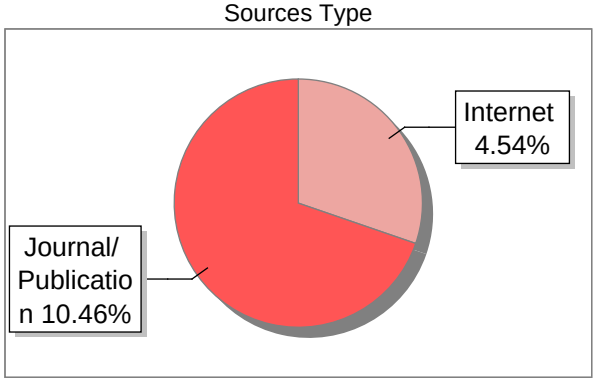
Author Name	manojkumar
Title	Coronary Artery Disease Detection
Paper/Submission ID	4870361
Submitted by	library@bmsce.ac.in
Submission Date	2025-12-10 14:56:40
Total Pages, Total Words	30, 4754
Document type	Project Work

Result Information

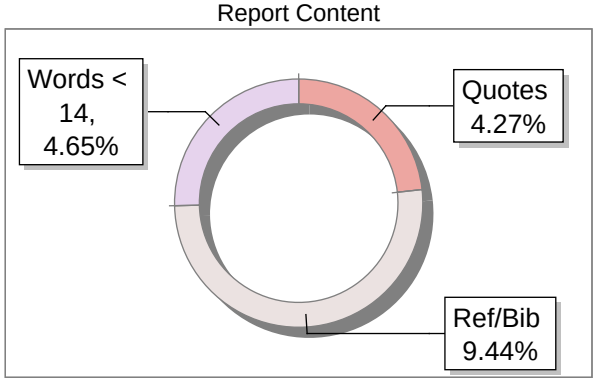
Similarity **15 %**

1 10 20 30 40 50 60 70 80 90

Sources Type



Report Content



Exclude Information

Quotes	Not Excluded
References/Bibliography	Not Excluded
Source: Excluded < 14 Words	Not Excluded
Excluded Source	**0 %**
Excluded Phrases	Not Excluded

Database Selection

Language	English
Student Papers	Yes
Journals & publishers	Yes
Internet or Web	Yes
Institution Repository	Yes

A Unique QR Code use to View/Download/Share Pdf File

DrillBit

DrillBit Similarity Report

15	39	B	A-Satisfactory (0-10%) B-Upgrade (11-40%) C-Poor (41-60%) D-Unacceptable (61-100%)
SIMILARITY %	MATCHED SOURCES	GRADE	

LOCATION	MATCHED DOMAIN	%	SOURCE TYPE
1	documents.newhorizonindia.edu	2	Publication
2	www.mdpi.com	1	Internet Data
3	arxiv.org	1	Publication
4	information-science-engineering.newhorizoncollegeofengineering.in	1	Publication
5	ieeexplore.ieee.org	1	Publication
6	www.scribd.com	1	Internet Data
7	translate.google.com	1	Internet Data
8	Optimization of multidimensional feature engineering and data partitioning stra, by Wang, Shanshan, Yr-2024	<1	Publication
9	ijsrst.com	1	Publication
10	sandbox.getindico.io	<1	Publication
11	link.springer.com	<1	Internet Data
12	arxiv.org	<1	Publication
13	hkbk.edu.in	<1	Publication
14	www.emerald.com	<1	Internet Data

15	pdfs.semanticscholar.org	<1	Publication
16	Deep learning-based CAD diagnosis using CNNs, by Amir Afzali, Mohsen, Yr-2025	<1	Publication
17	Parasitoid Coexistence Linking Spatial Field Patterns with Mechanism by Elizabet-2004	<1	Publication
18	scholarworks.rit.edu	<1	Publication
19	Thesis Submitted to Shodhganga Repository	<1	Publication
20	The burden of physical activity-related ill health in the UK by Allender-2007	<1	Publication
21	www.frontiersin.org	<1	Publication
22	www.ijitee.org	<1	Publication
23	www.kgcas.com	<1	Publication
24	openbioinformaticsjournal.com	<1	Publication
25	www.intechopen.com	<1	Internet Data
26	Assessing fairness in machine learning models A study of racial bias using matched counterparts in, by Wang, Yifei, Yr-2024	<1	Publication
27	cessi.iiserkol.ac.in	<1	Publication
28	docplayer.net	<1	Internet Data
29	egu.eu	<1	Internet Data
30	gnits.ac.in	<1	Internet Data
31	ieeexplore.ieee.org	<1	Publication
32	medium.com	<1	Internet Data